Predicting Apis Mellifera Pesticide Toxicity Using Machine Learning: A Study on the ApisTox Dataset

• Aaron Jeremy Alexander (2702238602) • Angela Kei Budiwinoto (2702309920) • Bella Nadya Aurelia (2702371565) • Ivana Joyceline (2702243092)

Problem Definition

Background:
Pesticide exposure has been a major cause of honeybee population decline. Understanding chemical toxicity to bees is crucial for ecological safety.
Objective:
This study aims to develop a machine learning model that accurately classifies the toxicity level (toxic or non-toxic) of chemical compounds to honeybees (Apis mellifera) based on molecular descriptors.

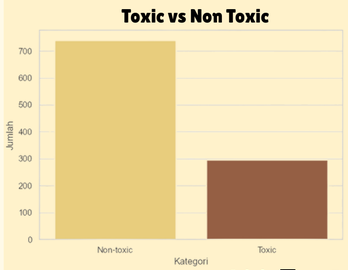
Data Collection and Data Description

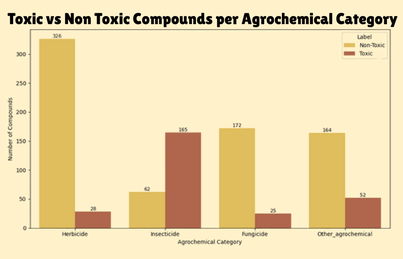
ApisTox **is a** dataset **centering** on the **toxicity** of pesticides to honeybees (Apis mellifera). This dataset may serve as a critical instrument for natural and agrarian research but can also bolster the advancement of approaches and hone in on minimizing harm to bee populations.

Data Prepocessing

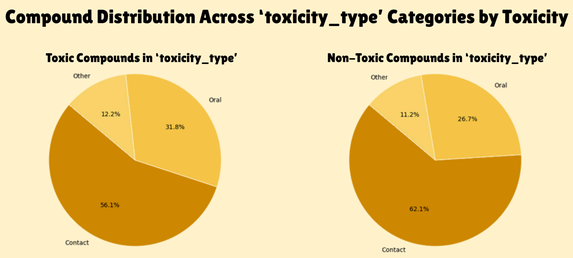
- Check for missing values and duplicate values (none found)

Exploratory Data Analysis (EDA)



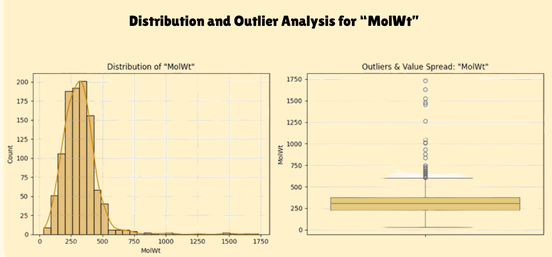


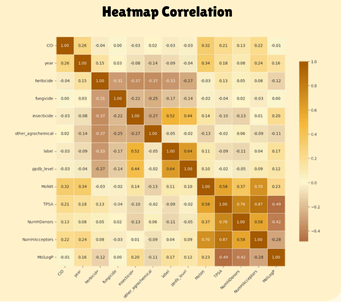
Label: Non Toxic Toxic



Feature Engineering and Selection

Turn SMILES column to features
SMILES was processed using RDKit to extract chemical decriptors such as MolWt, TPSA, MolLogP, NumHDonors, and NumHAcceptors. These features offer molecular-level information to support toxicity prediction.

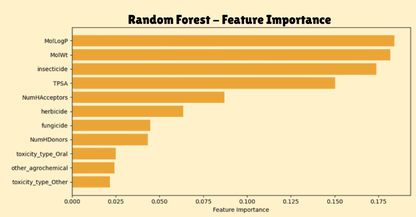


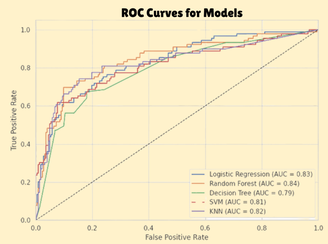


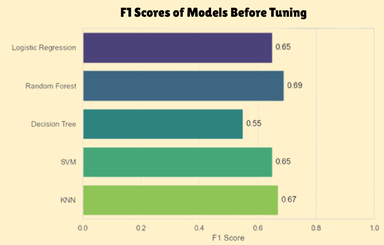
Model Selection and Training

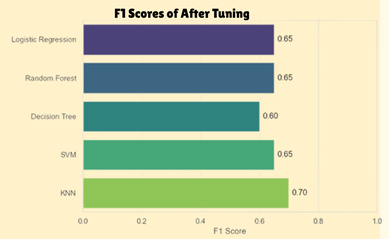
To identify the most effective approach for binary toxicity classification, we trained multiple models and ensemble methods—due to their strong performance in classification tasks. The dataset was split into 70% training and 30% testing for model evaluation.

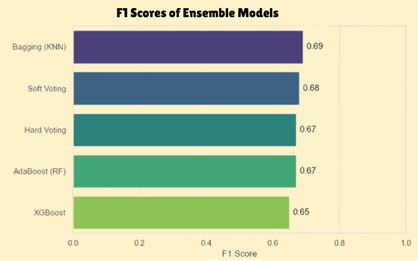
Model Evaluation and Hyperparameter Tunning











Comparison with Related Work

Study	Dataset	ML Method	Performance
Wang et al. (2020)	Public database, including Pesticide Properties Database by AERU, Pesticide Action Network Pesticide Database, ECOTOX Database by US EPA	Graph Attention Convolutional Neural Network (GAT-CNN)	The accuracy : 83.72% ~6% higher performance and more stable with ~7% Matthews Correlation Coefficient (MCC) higher compared to all tested models
Yang et al. (2022)	BeeToxAI dataset	SVM with two molecular representations : MACCS FP and L-RWGK	F1-Score MACCS FP vs L-RWGK : 0.7 vs 0.69
Our Study (2025)	ApisTox (UCI)	Logistic Regression, Random Forest, Decision Tree, SVM, KNN (all tuned)	F1-Score = 0.7 (KNN) AUC = 0.84 (Random Forest)

Discussion

This study applied various machine learning models to classify toxic chemical compounds using agrochemical attributes and molecular descriptors. Due to data imbalance, the F1 score was chosen as the main evaluation metric for its ability to balance precision and recall.
In terms of performance:

- Random forest achieved the highest F1 score in the baseline (0.69)
- After hyperparameter tuning, K-Nearest Neighbors (KNN) outperformed other models with score F1 = 0.70
- Bagging with KNN and Voting classifiers produced competitive results (Bagging F1 – 0.69)

Conclusion

Using molecular and agrochemical properties, machine learning algorithms may accurately forecast chemical toxicity. While Random Forest and Logistic Regression performed exceptionally well in AUROC, KNN obtained the best F1 score. The F1 score management for imbalance and SMILES feature engineering was crucial. All things considered, ML exhibits great promise for the early detection of toxic compounds.

References

- Adamczyk, J., Poziemski, J. & Siedlecki, P. ApisTox: a new benchmark dataset for the classification of small molecules toxicity on honey bees. Sci Data 12, 5 (2025). https://doi.org/10.1038/s41597-024-04232-w
- UCI Machine Learning Repository. (2023). ApiTox: A dataset for bee toxicity prediction. University of California, Irvine. https://archive.ics.uci.edu/dataset/995/apistox
- Yang, P., Henle, E. A., Fern, X. Z., & Simon, C. M. (2022). Classifying the toxicity of pesticides to honey bees via support vector machines with random walk graph kernels. The Journal of Chemical Physics, 157(3).
- Wang, F., Yang, J. F., Wang, M. Y., Jia, C. Y., Shi, X. X., Hao, G. F., & Yang, G. F. (2020). Graph attention convolutional neural network model for chemical poisoning of honey bees' prediction. Science Bulletin, 65(14), 1184-1191.